

Lemma: 群 G 作用在非空集 X 上. 对 $\forall x \in X$, 定义 x 的稳定化子为 $\text{Stab}_G(x) = \{g \in G \mid gx = x\} \leq G$. 则已经证明了 $\text{Stab}_G(x) \leq G$. 则 $G/\text{Stab}_G(x) = \{g \cdot \text{Stab}_G(x) \mid g \in G\}$.

定义映射 $f: G/\text{Stab}_G(x) \longrightarrow Gx$. 则映射 f 是双射.
$$g \text{Stab}_G(x) \longmapsto gx$$

Proof: 对 $\forall x \in X$, x 的 G -轨道为 $Gx = \{gx \mid g \in G\} \subseteq X$
 x 的稳定化子为 $\text{Stab}_G(x) = \{g \in G \mid gx = x\} \leq G$.

$$\therefore G/\text{Stab}_G(x) = \{g \text{Stab}_G(x) \mid g \in G\}.$$

任取 $G/\text{Stab}_G(x)$ 中的一元: $g \text{Stab}_G(x)$ (其中 $g \in G$)

$$\because g \in G \quad \because gx \in Gx \quad \therefore f(g \text{Stab}_G(x)) = gx \in Gx$$

任取 $G/\text{Stab}_G(x)$ 中的两个元素: $g_1 \text{Stab}_G(x), g_2 \text{Stab}_G(x)$
(其中 $g_1, g_2 \in G$)

若 $g_1 \text{Stab}_G(x) = g_2 \text{Stab}_G(x)$, 则 $g_1^{-1}g_2 \in \text{Stab}_G(x)$

$$\therefore g_1^{-1}g_2 \in G \text{ 且 } (g_1^{-1}g_2)x = x \quad \therefore g_1^{-1}(g_2x) = x.$$

$$\therefore g_1x = g_1(g_1^{-1}(g_2x)) = (g_1g_1^{-1})(g_2x) = 1_G(g_2x) = g_2x$$

$$\therefore f(g_1 \text{Stab}_G(x)) = g_1x = g_2x = f(g_2 \text{Stab}_G(x))$$

$\therefore f: G/\text{Stab}_G(x) \rightarrow Gx$ 是一个映射.

若 $f(g_1 \text{Stab}_G(x)) = f(g_2 \text{Stab}_G(x))$, 则 $g_1 x = g_2 x$

$$\therefore g_1^{-1}g_2 \in G \text{ 且 } (g_1^{-1}g_2)x = g_1^{-1}(g_2 x) = g_1^{-1}(g_1 x) = (g_1^{-1}g_1)x = 1_G x = x \quad \therefore g_1^{-1}g_2 \in \text{Stab}_G(x).$$

$\therefore G$ 是群, $g_1, g_2 \in G$, $\text{Stab}_G(x) \leq G$, $g_1^{-1}g_2 \in \text{Stab}_G(x)$

$$\therefore g_1 \text{Stab}_G(x) = g_2 \text{Stab}_G(x)$$

$\therefore f: G/\text{Stab}_G(x) \rightarrow Gx$ 是一个单射.

任取 Gx 中的一元 gx (其中 $g \in G$). $\therefore g \in G$ ~~$g \in \text{Stab}_G(x)$~~

$$\therefore g \text{Stab}_G(x) \in G/\text{Stab}_G(x)$$

$\therefore f(g \text{Stab}_G(x)) = gx \quad \therefore f: G/\text{Stab}_G(x) \rightarrow Gx$ 是一个满射.

$\therefore f: G/\text{Stab}_G(x) \rightarrow Gx$ 是一个双射. $\square$

Lemma: 群 G 作用在非空集 X 上. 对 $\forall g \in G$, $\forall x \in X$, 有:

$$\text{Stab}_G(gx) = g \text{Stab}_G(x) g^{-1}$$

Proof: $\therefore g \in G$ 且 $x \in X \quad \therefore gx \in X$

$$\therefore \text{Stab}_G(gx) = \{ \alpha \in G \mid \alpha(gx) = gx \} \leq G.$$

$$\therefore \text{Stab}_G(x) = \{ \beta \in G \mid \beta x = x \} \leq G, \quad g \in G \quad \therefore g \text{Stab}_G(x) g^{-1} \leq G$$

对 $\forall \alpha \in \text{Stab}_G(gx)$, 有: $\alpha \in G$ 且 $\alpha(gx) = gx$

$$\because g \in G \text{ 且 } \alpha \in G \quad \therefore g^{-1}\alpha g \in G$$

$$\therefore (g^{-1}\alpha g)x = g^{-1}((\alpha g)x) = g^{-1}(\alpha(gx)) = g^{-1}(gx)$$

$$= (g^{-1}g)x = 1_G x = x \quad \therefore g^{-1}\alpha g \in \text{Stab}_G(x)$$

$$\therefore g(g^{-1}\alpha g)g^{-1} \in g\text{Stab}_G(x)g^{-1} \quad \therefore \alpha \in g\text{Stab}_G(x)g^{-1}$$

$$\therefore \text{Stab}_G(gx) \subseteq g\text{Stab}_G(x)g^{-1}$$

任取 $g\text{Stab}_G(x)g^{-1}$ 中的元: $g\beta g^{-1}$ (其中 $\beta \in \text{Stab}_G(x)$)

$$\because \beta \in \text{Stab}_G(x) \quad \therefore \beta \in G \text{ 且 } \beta x = x \quad \therefore g \in G \text{ 且 } \beta \in G$$

$$\therefore g\beta g^{-1} \in G \quad \therefore (g\beta g^{-1})(gx) = (g\beta)(g^{-1}(gx)) = (g\beta)((g^{-1}g)x)$$

$$= (g\beta)(1_G x) = (g\beta)x = g(\beta x) = gx$$

$$\therefore g\beta g^{-1} \in \text{Stab}_G(gx) \quad \therefore g\text{Stab}_G(x)g^{-1} \subseteq \text{Stab}_G(gx)$$

$$\therefore \text{Stab}_G(gx) = g\text{Stab}_G(x)g^{-1} \quad \square$$

Lemma: 群 G 作用在有限非空集 X 上. 则已经证明了 X 是其所有

G -轨道的无交并. 设 $Gx_1 \subseteq X$, $Gx_2 \subseteq X$, $\dots$, $Gx_n \subseteq X$

(其中 $n \in \mathbb{Z}_{\geq 1}$, $x_1, \dots, x_n \in X$, 对 $\forall i, j \in \{1, \dots, n\}$ 且 $i \neq j$, 有

$$Gx_i \cap Gx_j = \emptyset). \text{ 则 } X = \bigsqcup_{i=1}^n Gx_i$$

$$\therefore |X| = \sum_{i=1}^n |Gx_i| = \sum_{i=1}^n |G/\text{Stab}_G(x_i)| = \sum_{i=1}^n (G : \text{Stab}_G(x_i)) \quad \square$$

Burnside 3|理: 有限群 G 作用于有限非空集 X . 对 $\forall g \in G$ 定义 $X^g := \{x \in X \mid gx = x\}$. 则有 $|G \backslash X| \cdot |G| = \sum_{g \in G} |X^g|$

Proof: $\because G$ 是有限群 $\therefore$ 设 $G = \{g_1, \dots, g_n\}$ (其中 $n \in \mathbb{Z}_{\geq 1}$)

$$\therefore |G| = n \in \mathbb{Z}_{\geq 1}$$

$\because X$ 是有限非空集合 $\therefore$ 设 $X = \{x_1, \dots, x_m\}$ (其中 $m \in \mathbb{Z}_{\geq 1}$)

$$\therefore |X| = m \in \mathbb{Z}_{\geq 1}$$

定义矩阵 $A = (a_{ij})_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}} \in M_{n \times m}(\mathbb{R})$ 为: 对 $\forall i \in \{1, \dots, n\}$,

$$\forall j \in \{1, \dots, m\}, \quad a_{ij} = \begin{cases} 1 & , \text{若 } g_i x_j = x_j \\ 0 & , \text{若 } g_i x_j \neq x_j \end{cases}$$

$$\therefore \text{矩阵 } A \text{ 中所有元素的和} = \sum_{i=1}^n \sum_{j=1}^m a_{ij} = \sum_{i=1}^n \left(\sum_{j=1}^m a_{ij} \right)$$

对 $\forall i \in \{1, \dots, n\}$, 有: $g_i \in G$

$$\therefore a_{ij} = 1 \iff g_i x_j = x_j \iff x_j \in X^{g_i}$$

$$\therefore \text{矩阵 } A \text{ 中所有元素的和} = \sum_{i=1}^n \left(\sum_{j=1}^m a_{ij} \right) = \sum_{i=1}^n |X^{g_i}|$$

$$\therefore \text{矩阵 } A \text{ 中所有元素的和} = \sum_{j=1}^m \left(\sum_{i=1}^n a_{ij} \right)$$

对 $\forall j \in \{1, \dots, m\}$, 有: $x_j \in X$

$$\therefore a_{ij} = 1 \iff g_i x_j = x_j \iff g_i \in \text{Stab}_G(x_j)$$

$$\therefore \text{矩阵 } A \text{ 中所有元素的和} = \sum_{j=1}^m \left(\sum_{i=1}^n a_{ij} \right) = \sum_{j=1}^m |Stab_G(x_j)|$$

$$\therefore \sum_{i=1}^n |X^{g_i}| = \sum_{j=1}^m |Stab_G(x_j)|$$

对 $\forall j \in [1, \dots, m]$, 有: $x_j \in X \quad \therefore Stab_G(x_j) \leq G$, 且有

$$|G/Stab_G(x_j)| = |Gx_j|$$

$\therefore G$ 是有限群, $Stab_G(x_j) \leq G \quad \therefore |Stab_G(x_j)| \cdot (G:Stab_G(x_j)) = |G|$

$|G| = n \in \mathbb{Z}_{\geq 1}$, $|Stab_G(x_j)| \in \mathbb{Z}_{\geq 1}$, $\therefore (G:Stab_G(x_j)) \in \mathbb{Z}_{\geq 1}$

$$\therefore |Stab_G(x_j)| = \frac{|G|}{(G:Stab_G(x_j))}$$

$$\therefore \sum_{i=1}^n |X^{g_i}| = \sum_{j=1}^m |Stab_G(x_j)| = \sum_{j=1}^m \frac{|G|}{(G:Stab_G(x_j))}$$

$$= |G| \sum_{j=1}^m \frac{1}{(G:Stab_G(x_j))} = |G| \sum_{j=1}^m \frac{1}{|G/Stab_G(x_j)|}$$

$$= |G| \sum_{j=1}^m \frac{1}{|Gx_j|}$$

$\therefore$ 集合 X 是其所有 G -轨道的无交并. 设 X 的所有 G -轨道为

$$O_1, \dots, O_k \quad (\text{其中 } k \in \mathbb{Z}_{\geq 1}) \quad \therefore X = \bigsqcup_{r=1}^k O_r$$

$$\therefore G \backslash X = \{O_1, \dots, O_k\} \quad \therefore |G \backslash X| = k \in \mathbb{Z}_{\geq 1}$$

$$\therefore \sum_{i=1}^n |X^{g_i}| = |G| \sum_{j=1}^m \frac{1}{|Gx_j|} = |G| \sum_{x \in X} \frac{1}{|Gx|}$$

$$= |G| \sum_{r=1}^k \left(\sum_{x \in O_r} \frac{1}{|Gx|} \right) = |G| \cdot \sum_{r=1}^k \left(\sum_{x \in O_r} \frac{1}{|O_r|} \right)$$

$$= |G| \cdot \sum_{r=1}^k \left(|O_r| \cdot \frac{1}{|O_r|} \right) = |G| \cdot \sum_{r=1}^k 1 = k \cdot |G|$$

$$= |G \setminus X| \cdot |G|$$

$$\therefore |G \setminus X| \cdot |G| = \sum_{j=1}^n |X^{g_j}| = \sum_{g \in G} |X^g| \quad \square$$

Remark: 有限群 G 作用于有限非空集 X . 则有: $|G| \in \mathbb{Z}_{\geq 1}$,
 $|X| \in \mathbb{Z}_{\geq 1} \because X \neq \emptyset \therefore X$ 至少有 1 个 G -轨道.

$\therefore G \setminus X \neq \emptyset \because X$ 是有限集 $\therefore X$ 只有有限多个 G -轨道

$$\therefore |G \setminus X| \in \mathbb{Z}_{\geq 1}$$

$\therefore 1_G \in G$, 且对 $\forall x \in X$, 都有 $1_G x = x$

$\therefore$ 对 $\forall x \in X$, 都有 $x \in X^{1_G} \therefore X^{1_G} = X \therefore |X^{1_G}| = |X| \in \mathbb{Z}_{\geq 1}$

$\therefore$ 对 $\forall g \in G$, $X^g \subseteq X \therefore$ 对 $\forall g \in G$, $|X^g| \in \mathbb{Z}_{\geq 0}$

$$\therefore |G \setminus X| = \frac{1}{|G|} \cdot \sum_{g \in G} |X^g|$$